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SUMMARY OF QUALIFICATIONS

- 4+ years of software development experience with a focus on object-oriented programming, including 16+ months of hands-on, paid co-op experience across four organizations developing and consuming RESTful APIs and building full-stack, backend, and embedded applications in Agile/Scrum cross-functional teams
- Proficient in C#/.NET (Blazor, MudBlazor, Entity Framework) and object-oriented design principles, with additional experience in JavaScript, TypeScript, React.js/Redux, Node.js, Express.js, Java, Python, C, and C++; builds and validates REST APIs with MySQL, MongoDB, and PostgreSQL; unit, integration, and BDD testing (Gherkin)
- Working familiarity with Microsoft Azure and AWS cloud services and deployment strategies, plus Docker, CI/CD, Git/GitHub/GitLab, Jira, Confluence, and Postman; demonstrated experience collaborating within a development team throughout the Agile Scrum product development lifecycle
- Hands-on experience building AI agents and daily use of AI development tools (GitHub Copilot, ChatGPT, Claude API/Anthropic SDK, Vercel AI SDK, Google Gemini API, OpenCode) to improve productivity and accelerate delivery
- Bachelor's degree in software engineering (Bilingual English/French), University of Ottawa Co-op, CGPA 9.7/10, Dean's Honor Award every term; Fortinet NSE 4 (FortiOS 7.6) certified; strong analytical, problem-solving, and communication skills built through teaching, mentoring, and team-based project work; proactive, growth-oriented attitude toward learning and professional development

EDUCATION

Bachelor of Applied Science, Software Engineering (Co-op)

2022 – 2026

University of Ottawa, Ottawa, ON

- French Immersion Stream and Engineering Management & Entrepreneurship option | CGPA: 9.7/10 | Dean's Honor Award & Merit Scholarship: All terms
- Relevant courses: Real-Time & Embedded Software Design, Secure Enterprise Network Infrastructure (Fortinet), Data Communications & Networking (Cisco), Software Engineering Capstone, Software Design & Architecture, Software Quality Assurance, Database, Operating Systems, Analysis & Design of User Interfaces

RELEVANT EXPERIENCE

Software Developer Intern

May – Aug 2025

University of Ottawa – Co-op Department, Ottawa, ON

- Served as the second developer on uOCompetencies, a newly launched web application enabling co-op students to build digital learning portfolios, set competency development goals, and receive structured mentor feedback across Work-Study, Self-Directed, Internship, and Short-Term Activity experiential learning types
- Developed and maintained new frontend and backend features using the Microsoft Blazor framework, MudBlazor components, and C#, applying object-oriented programming principles to translate product requirements into functional, testable, and maintainable code on a still-evolving codebase, improving system functionality and usability for administrative staff
- Debugged, tested, and resolved software defects across InDEX — the University's internal co-op job board and database, functioning similarly to Indeed and used by 4,000+ co-op students annually to discover placements outside standard COOP Navigator postings and timelines — and the notificationSender App, by adapting existing C#, HTML, CSS, and VB.net code to new requirements
- Collaborated with the development team via GitLab using Agile practices (sprint planning, issue tracking, peer code review) to ensure timely, high-quality delivery across both applications

Software Developer Intern

Sep – Dec 2024

Warner Bros. Discovery, Ottawa, ON

- Wrote clean, testable, object-oriented code to develop and maintain new features, resolve bugs, and write unit tests for the Max streaming application on Roku devices, applying BrightScript, SceneGraph, and XML to improve application reliability and maintainability for a large subscriber base
- Executed test cases against assigned Jira tickets during a dedicated biweekly TestFest phase to verify bug fixes and feature changes prior to release, and documented test outcomes and defect resolutions in Confluence
- Participated in code review, debugging, and troubleshooting by critically analyzing pull requests from cross-functional teammates during MergeFest, helping uphold code quality and test coverage standards before merge
- Tracked tasks, defects, and sprint progress using Jira and Confluence within a biweekly Agile cycle comprising MergeFest, TestFest, and TeamFest

Web Developer Intern

Jan – Apr 2024

University of Ottawa – Teaching and Learning Support Service (TLSS), Ottawa, ON

- Built frontend components for BrightSpace e-learning courses from storyboards using HTML, CSS, JavaScript, jQuery, and Bootstrap, supporting course content delivered to a large base of enrolled students
- Collaborated with instructional designers to produce accessible, user-friendly interfaces that met diverse learner needs and university accessibility standards

IT Support Analyst Intern & Part-Time

May 2023 – Apr 2025

University of Ottawa – Central IT Department, Ottawa, ON

- Diagnosed and resolved software and hardware issues for students and staff by configuring MFA, VPN, VMware, and cloud backups, maintaining reliable access to university IT services
- Managed device imaging via SCCM and administered virtual machines with Hyper-V, supporting consistent, reliable environment setup and configuration across university IT infrastructure

Teaching Assistant / Lab Demonstrator / Corrector – ITI 1100

Jan – Apr 2026

University of Ottawa, Faculty of Engineering, Ottawa, ON

- Guided students through weekly lab sessions and course material, clarifying complex programming and digital systems concepts for a first-year engineering course
- Evaluated assignments, midterms, and exams with constructive feedback, applying strong written and verbal communication skills

TECHNICAL SKILLS

Programming Languages (OOP): C#, Java, TypeScript (ES6), JavaScript, Python, C, C++, BrightScript, Visual Basic, PHP

Full-Stack, .NET & APIs: .NET, Blazor, Entity Framework, RESTful API design & consumption, React.js/Redux, Node.js, Express.js, HTML5, CSS3/SASS, Bootstrap, jQuery, Axios

Cloud & Dev Tools: Microsoft Azure (familiar), AWS (familiar), Docker, CI/CD, Git, GitHub, GitLab, Jira, Confluence, Postman, SCCM, Hyper-V, VMware, node-cron

Agile & Collaboration: Agile/Scrum (sprint planning, daily stand-ups, retrospectives, code review), cross-functional team collaboration, technical documentation

Databases: MySQL, MongoDB, PostgreSQL, MS SQL Server, Firebase

AI & Modern Workflows: GitHub Copilot, ChatGPT, Claude API/Anthropic SDK, Vercel AI SDK, Google Gemini API, Agentic AI Systems/LLM Tool-Calling, Playwright Browser Automation, OpenCode, Prompt Engineering, RAG

Testing & QA: TDD, BDD, Gherkin, unit & integration testing, API testing (Postman), defect tracking (Jira), Espresso

Networking & Security: Fortinet NSE 4 (FortiOS 7.6), Cisco networking, VPN, MFA, cross-platform desktop (Windows, macOS, Linux)

APPLIED PROJECTS

Autonomous AI Web-Form Agent – Magical LLM Take-Home Challenge

Jul 2026

- Built an autonomous AI agent with the Vercel AI SDK and Google Gemini that completes a multi-section healthcare web form via Playwright browser automation, implementing 8 custom tools (click, fill, select, scroll) in an agentic tool-calling loop with retry/exponential-backoff and automatic page-reload error handling
- Extended the agent to run as a REST API (Express.js) with a POST /workflow endpoint, CLI/environment-variable input for dynamic form data, and a node-cron scheduler that triggers automated runs every 5 minutes with overlap protection

IoT Security Testbed (IoTa) – Software Engineering Capstone

Sep 2025 – Apr 2026

- Designed and implemented an IoT security testbed by generating datasets across four protocols (Zigbee, MQTT, Thread, and Matter); built automated logging APIs and services (TimescaleDB, Redis, BullMQ, Express.js) and a Grafana dashboard to visualize and analyze security events in real time
- Explored and implemented machine learning anomaly detection on generated IoT datasets, using AI tools (GitHub Copilot, OpenCode) to accelerate model experimentation and ML pipeline development

Full-Stack Hotel Management System

Feb – Apr 2025

- Collaborated with a team of 4 to build a hotel management system using React.js, Node.js, Express.js, and PostgreSQL; connected frontend to backend REST APIs using Axios and CORS
- Wrote and executed Postman API test scripts to validate all endpoints, ensuring system reliability prior to integration

Full-Stack Pharmacy Prescription Management System (UYP-PMS)

Sep 2025

- Developed in a team of 5 using Spring Boot (Kotlin), Angular, and PostgreSQL; implemented role-based authentication with Spring Security and containerized the application with Docker
- Validated end-to-end user workflows using BDD test procedures written in Gherkin to confirm functional requirements were met

Ottawa Badminton Club – Responsive Website

Jun – Jul 2024

- Designed and developed a fully responsive website using React.js, Node.js, Bootstrap, HTML, CSS, SASS, JavaScript, jQuery, JSON, and XML; applied user-centered design and HCI prototyping best practices
- Achieved a 100% Google Lighthouse accessibility score by applying heuristic evaluation, WCAG-compliant design, and user goal/task analysis to deliver an inclusive, intuitive interface

EXTRA-CURRICULAR ACTIVITIES AND VOLUNTEER EXPERIENCE

- **Hackathon Participant | uOttawa** — Collaborated with peers to develop software solutions under time constraints
- **Engineering Guide (101 Week) | Software Engineering Student Association (uOttawa SESA), Sep 2025/2026** — Guided first-year students and completed training in first aid, sexual violence awareness, and student support
- **Cook | PRD_uOttawa Club, Sep – Dec 2022** — Assisted in preparing free plant-based meals from would-be food waste every week at university
- **Member | High School UNESCO Club, Jan – Dec 2020** — Supported an anti-drug campaign, tutored English, French, and Math, and organized activities at an elderly care home